

FRC JAVA CODING LAB 7.0

OBSERVATION ARCHITECTURE CHECKLIST

Mandatory review for `frc.robot.observation`

Document	Version	Status	Language
OC-03	1.0	FROZEN	English

SSIS FRC Team 10951

Author: SSIS | Mentor: SSIS

Review Information

Project: _____ Lesson/Module: _____
 Reviewer: _____ Date: _____
 Result: PASS / FAIL / REQUIRES MIGRATION

A. Frozen Flows

- ☐ Control remains Driver -> Xbox Controller -> controls -> commands -> subsystems -> io -> hardware.
- ☐ Observation remains hardware -> IOInputs -> subsystem/estimator -> immutable Observation -> telemetry -> NT4/Glass/log.
- ☐ Observation cannot control, schedule, configure, or write hardware.

B. Ownership and Dependencies

- ☐ IO owns hardware access and the mutable one-cycle IOInputs snapshot.
- ☐ Subsystem or estimator produces the Observation.
- ☐ Telemetry only consumes and publishes Observations.
- ☐ Observation has no vendor, hardware, NetworkTables, CommandScheduler, RobotContainer, command, control, or telemetry dependency.
- ☐ util contains only genuinely shared helpers; mechanism observation meaning remains in observation.

C. Model Contract

- ☐ Observation is immutable and does not retain mutable aliases.
- ☐ Type and component names state responsibility.
- ☐ Units are explicit in names or JavaDoc.
- ☐ Timing basis and sample coherence are defined when time matters.
- ☐ Validity, disconnection, staleness, unsupported values, and estimated values are explicit when applicable.
- ☐ No invalid value silently appears as a valid zero.

D. Evaluators

- ☐ Evaluators are pure, stateless, deterministic, and side-effect free.
- ☐ All inputs and configuration are explicit.
- ☐ No clock, singleton, supplier side effect, cache, publisher, hardware, or runtime environment is read.
- ☐ Boundary and repeatability tests exist.

E. Telemetry and Logging

- ☐ Telemetry owns topic names, publisher lifecycle, rate, and serialization.
- ☐ Typed stable topics are used.
- ☐ Telemetry performs no behavior control and does not mutate observations.
- ☐ Logging consumes the same immutable meaning or a documented compatible projection.

F. L10 and Migration Review

- ☐ L10 examples were reviewed but not treated as authority.
- ☐ Existing immutable records and pure evaluator patterns align.
- ☐ Direct IOInputs publication, missing required validity/time, mutable models, or stateful evaluators are recorded for migration.
- ☐ Completed lessons remain unchanged; migration is planned for the next editable lesson.

G. Document and Release Review

- ☐ Document A, ES-06, Document B, AGENTS.md, README.md, hardware ZIP, L10 ZIP/source were reviewed.
- ☐ EN is normative and VI is explanatory with matching structure and meaning.
- ☐ Version 1.0, FROZEN, Author SSIS, and Mentor SSIS are present.
- ☐ DOCX and PDF render without clipping, overlap, broken tables, or missing glyphs.

Decision Record

Conflicts: _____
Required migration: _____
Evidence: _____
Final decision: PASS / FAIL / REQUIRES MIGRATION

Revision History

1.0 | 2026-08-01 | FROZEN | Initial release